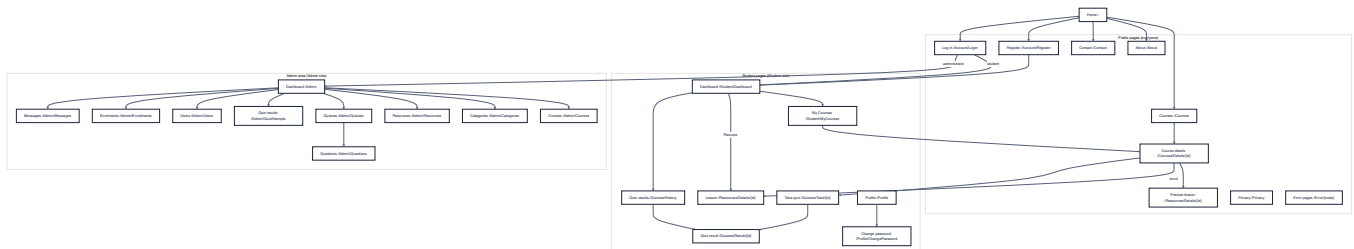


Navigation Structure

Navigation is defined in `src/LearnHub/Views/Shared/_Layout.cshtml` (public and student pages) and `src/LearnHub/Areas/Admin/Views/Shared/_AdminLayout.cshtml` (admin area). The current section is highlighted by the `lh-nav` tag helper, which also sets `aria-current="page"`.

1. Site map



2. Main navigation by role

Element	Guest	Student	Administrator
Brand (LearnHub mark)	Home	Home	Home
Home, Courses	Yes	Yes	Yes
My Courses, Dashboard	–	Yes	–
About, Contact	Yes	In the footer	In the footer
Admin	–	–	Yes (opens <code>/Admin</code>)
Right-hand side	Log in, Register	User menu: Dashboard, Quiz results, Profile, Log out	User menu: Admin dashboard, Profile, Log out

Log out is a POST form inside the user menu, so it cannot be triggered by a link on another site.

Admin area sidebar

Group	Links
–	Dashboard
Catalogue	Courses, Categories, Resources
Assessment	Quizzes (also highlighted on question pages), Quiz results
People	Users, Enrolments, Messages (with the unread count)
Website	Main website, Log out

The admin top bar shows the page title, **View site** and the administrator's profile link.

Footer

Column	Links
Explore	All courses, About LearnHub, Contact
Account	Guest: Create an account, Log in, Privacy. Signed in: Profile, Privacy
Project	Module and university information

3. Breadcrumbs

Breadcrumbs appear where pages are nested more than one level deep:

- **Public and student:** course catalogue and course details (Home / Courses / category / course), lessons (Courses / course / lesson), quiz and quiz result pages, and Change password.
- **Admin:** create, edit, details and delete pages for courses, categories, resources, quizzes, questions, quiz attempts, enrolments, users and messages, for example Courses / *course title* / Edit.

4. Routes

URL	Page	Access	Methods
/	Home	Everyone	GET
/About , /Privacy	About, Privacy	Everyone	GET
/Contact	Contact form	Everyone (rate limited)	GET, POST
/Courses? q=&categoryId=&difficulty=&sort=&page=	Course catalogue	Everyone	GET
/Courses/Details/{id}	Course details	Everyone (published courses)	GET
/Courses/Enroll/{id} , /Courses/Leave/{id}	Enrol, leave	Student	POST
/Resources/Details/{id}	Lesson	Preview: everyone; otherwise enrolled students and administrators	GET
/Resources/Open/{id}	Lesson file (PDF or image)	Same as the lesson	GET
/Resources/ToggleComplete/{id}	Mark complete or not complete	Enrolled student	POST
/Quizzes/Take/{id}	Quiz form and submission	Enrolled student	GET, POST

URL	Page	Access	Methods
/Quizzes/Result/{id}	Quiz result	The student who submitted it	GET
/Quizzes/History	Quiz results history	Student	GET
/Student/Dashboard , /Student/MyCourses	Dashboard, My Courses	Student	GET
/Profile	Profile	Signed-in user	GET, POST
/Profile/ChangePassword	Change password	Signed-in user (rate limited)	GET, POST
/Account/Register , /Account/Login	Register, log in	Everyone (rate limited)	GET, POST
/Account/Logout	Log out	Signed-in user	POST
/Account/AccessDenied	Access denied (status 403)	Everyone	GET
/Error , /Error/{statusCode}	Friendly error pages	Everyone	GET
/health	Database health check	Everyone	GET
/Admin	Admin dashboard	Admin	GET
/Admin/Courses + /Create , /Edit/{id} , /Details/{id} , /Delete/{id} , /TogglePublished/{id}	Course management	Admin	GET, POST
/Admin/Categories + /Create , /Edit/{id} , /Delete/{id}	Category management	Admin	GET, POST
/Admin/Resources + /Create , /Edit/{id} , /Delete/{id}	Learning resource management	Admin	GET, POST
/Admin/Quizzes + /Create , /Edit/{id} , /Details/{id} , /Delete/{id} , /SetPublished/{id}	Quiz management	Admin	GET, POST
/Admin/Questions/Create?quizId= + /Edit/{id} , /Delete/{id}	Question management	Admin	GET, POST
/Admin/QuizAttempts + /Details/{id} , /Delete/{id}	Quiz results	Admin	GET, POST

URL	Page	Access	Methods
/Admin/Enrollments + /Create , /Delete/{id}	Enrolment management	Admin	GET, POST
/Admin/Users + /Details/{id} , /SetRole/{id} , /Deactivate/{id} , /Reactivate/{id} , /Delete/{id}	User management	Admin	GET, POST
/Admin/Messages + /Details/{id} , /MarkUnread/{id} , /Delete/{id}	Contact messages	Admin	GET, POST

Every state-changing action is a POST protected by an antiforgery token.

5. Redirects and error behaviour

Situation	Result
A guest opens a page that needs sign-in	302 to /Account/Login?ReturnUrl=... ; after login the user returns to that page if it is on this site
A signed-in user opens a page for another role	302 to /Account/AccessDenied , which responds with status 403
An unknown URL, a draft course or a non-existent id	Friendly 404 page ("This stop is not on the map") with links back to courses
Another student's quiz result	404, so result ids cannot be probed
A student opens a lesson or quiz of a course they have not joined	Redirect to the course page, which offers Enrol
An unexpected error in production	Friendly 500 page without technical details; the error is logged
Too many form submissions	429 Too Many Requests

6. Responsive behaviour

- **Main navigation:** below 992 px the menu collapses behind a toggle button labelled "Toggle navigation".
- **Admin sidebar:** below 992 px it becomes an off-canvas panel opened from the top bar ("Open navigation").
- **Admin tables:** below 768 px they become stacked records, each value shown with its column label; wider tables scroll inside their own container.
- **Course grids:** one column on phones, two on tablets, and three or four on desktops.
- **Skip links:** every layout starts with "Skip to main content" for keyboard users.